

TreeDiff: AST-Guided Code Generation with Diffusion-based LLMs (DLLMs)

Yiming Zeng^{1*} · Jinghan Cao^{2*} · Zexin Li³ · Yiming Chen⁴ · Tao Ren⁵ · Zhuochun Li⁵ · Dawei Xiang¹ · Xidong Wu⁵ · Shangqian Gao⁶ · Tingting Yu¹

¹University of Connecticut · ²San Francisco State University · ³UC Riverside · ⁴National University of Singapore · ⁵University of Pittsburgh · ⁶Florida State University · * Equal contribution

 **TreeDiff** = first AST-aware masking for diffusion code generation → **+13.3% on HumanEval+**, narrows gap with autoregressive models.

Motivation

Diffusion-based LLMs (DLLMs) struggle with **syntactic precision** in code generation. Standard random token masking:

- Ignores syntactic boundaries during denoising
- Fails to capture long-range hierarchical dependencies
- Treats all tokens equally — but they're not!

CORE INSIGHT

"For code, noise should not be purely stochastic — structural priors enable DLLMs to learn program-level dependencies."

Method: AST-Guided Masking

TreeDiff parses code into an **Abstract Syntax Tree (AST)** and uses it to guide which tokens to mask — targeting structurally critical nodes.

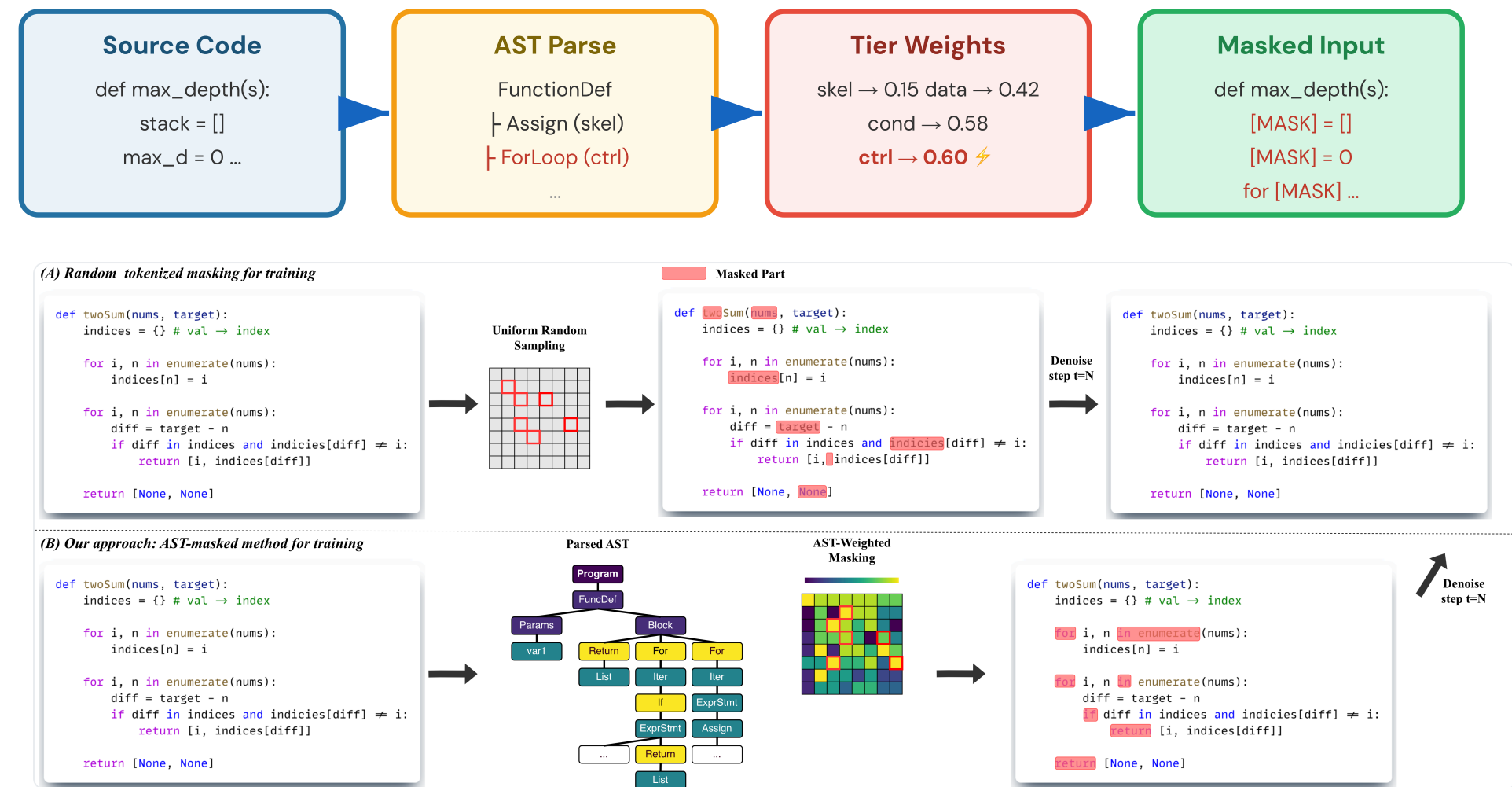
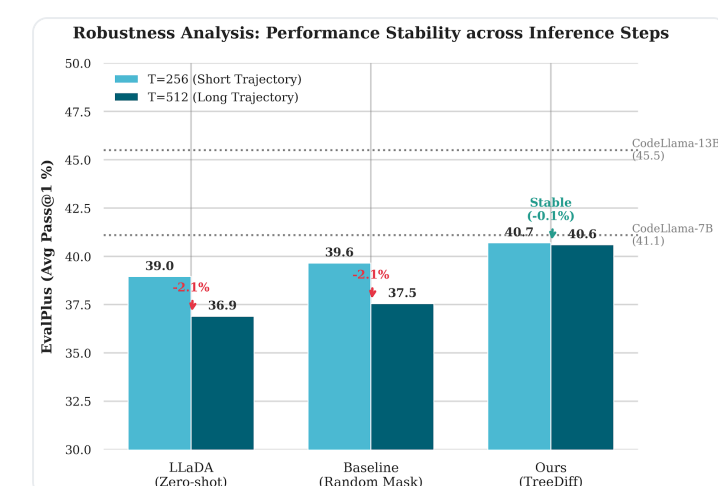


Figure 1: Uniform Random Masking (top) vs. AST-Weighted Sampling Masking (bottom).

AST Node Weighting

Category	Examples	Weight	Purpose
Skeletal	Imports, Constants	0.15	Preserve skeleton
Data Flow	Assigns, Calls	0.42	Moderate priority
Conditionals	if, elif, else	0.58	Core logic
Control Flow	for, while, return	0.60	⚡ Highest priority

Robustness Analysis



TreeDiff maintains stable performance (−0.1%) as T increases 256→512, while baselines degrade by ~2.1%.

Main Results (Pass@1 %)

Model	T = 256				T = 512			
	HE	HE+	MBPP	MBPP+	HE	HE+	MBPP	MBPP+
Autoregressive Baselines								
DeepSee... Coder-33B	50.6	44.5	80.4	70.1	50.6	44.5	80.4	70.1
CodeQw... 7B	51.8	45.7	73.5	60.8	51.8	45.7	73.5	60.8
CodeLla... 13B	42.7	38.4	63.5	52.6	42.7	38.4	63.5	52.6
Diffusion-based LLMs (LLaDA Backbone)								
LLaDA-Instruct	39.6	34.8	51.9	43.1	36.6	31.7	39.4	31.7
+ Random Masking	39.6	35.4	52.9	43.9	38.4	32.3	41.5	32.8
Ours								
+ TreeDiff	42.1	37.2	52.9	44.2	40.2	36.6	42.3	33.3

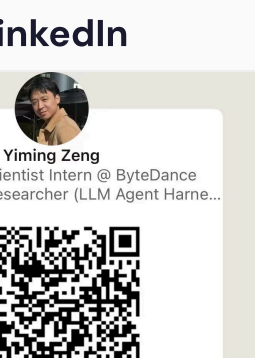
Source: AR baselines from EvalPlus Leaderboard. HE = HumanEval; HE+ = HumanEval+.

Ablation: Masking Granularity

HumanEval		
Strategy	T=256	T=512
Random Masking	39.6 / 35.4	38.4 / 32.3
AST (Span)	40.9 / 36.6	40.2 / 36.6
AST (Node)	42.1 / 37.2	40.2 / 36.6
MBPP		
Strategy	T=256	T=512
Random Masking	52.9 / 43.9	41.5 / 32.8
AST (Span)	51.6 / 42.9	39.7 / 31.5
AST (Node)	52.9 / 44.9	42.3 / 33.3

Key finding: Span-level masking corrupts entire subtrees — too coarse. TreeDiff's **token-level** AST masking surgically targets high-influence nodes while preserving context.

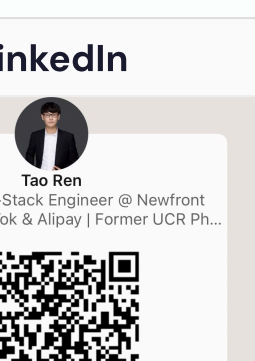
Contact Authors



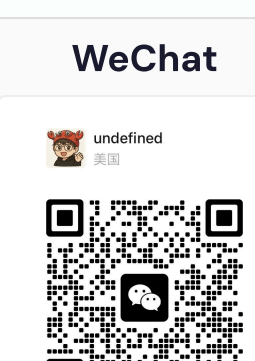
Yiming Zeng
yiming.zeng@uconn.edu



WeChat



Tao Ren
tar118@pitt.edu



WeChat



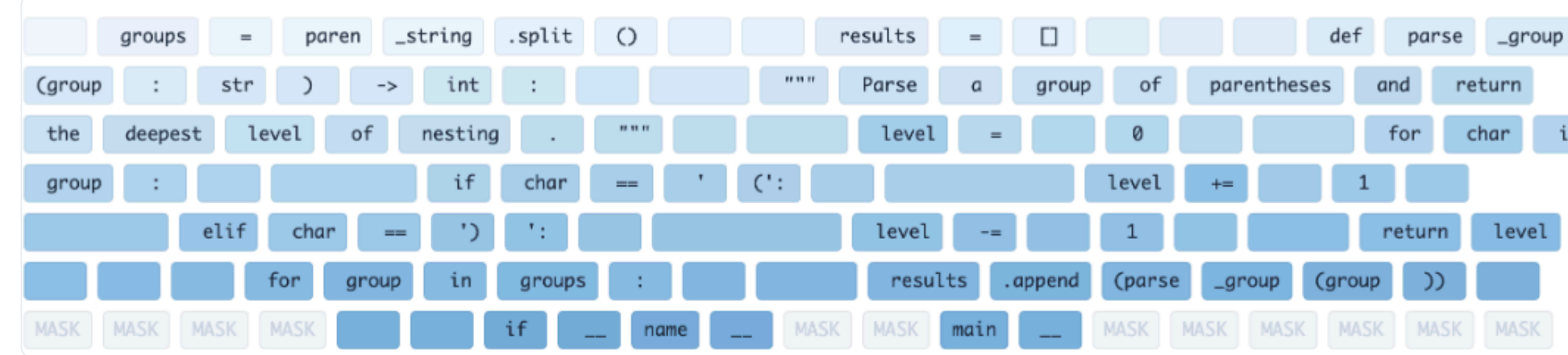
Zexin Li
zli536@ucr.edu

Website

We welcome collaboration and discussion. Feel free to reach out!

Generation Trajectory Analysis

Task: Max nesting depth of parentheses (HumanEval/6), Step 110/256.



Random Mask Baseline: Uses shallow counter heuristic, fails on nested structures. Fragmented generation.



TreeDiff: Establishes stack-based control flow (stack.pop()) early. Correctly captures recursive nesting.

<> Generated Code Comparison

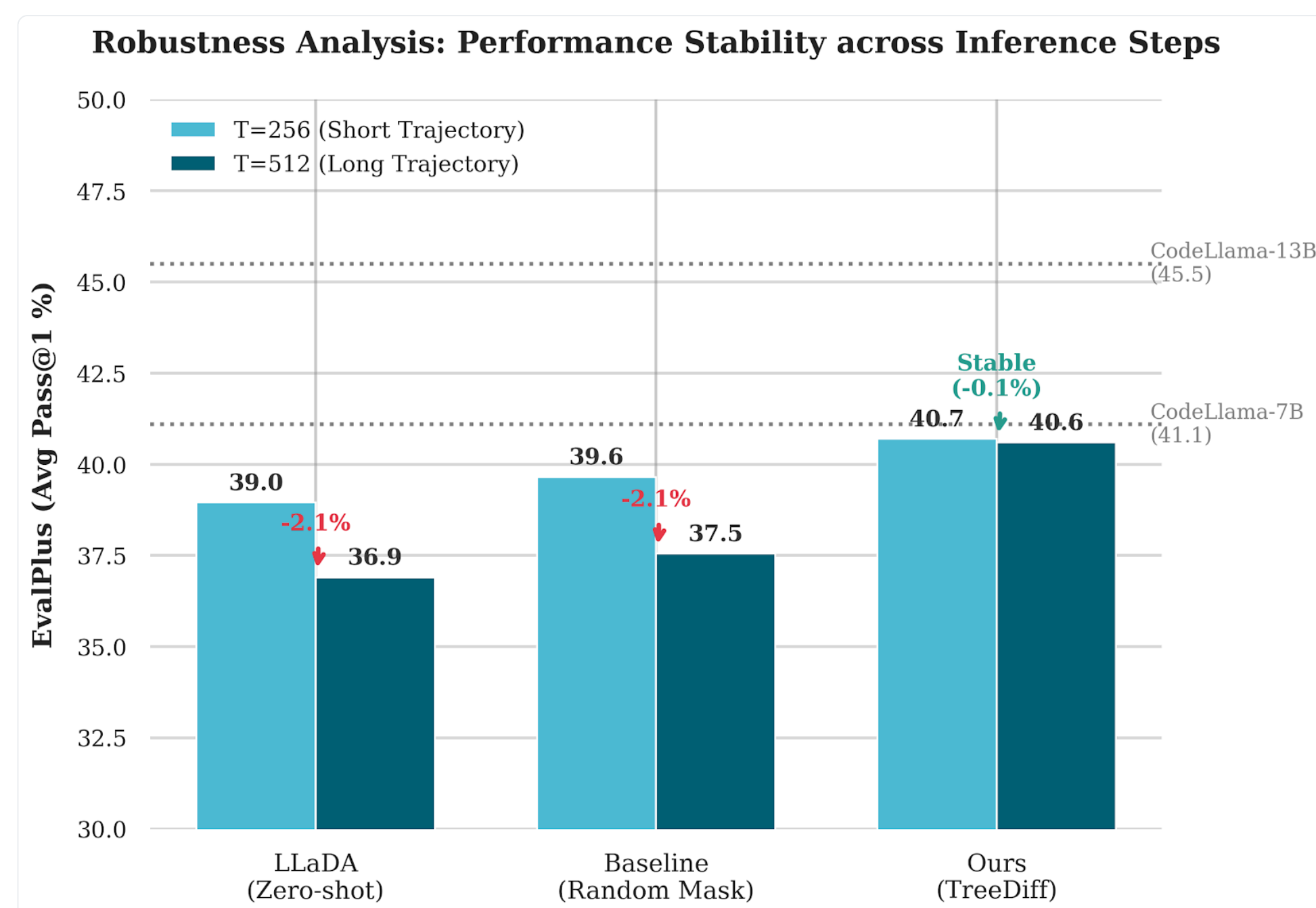
HumanEval/6 — Nested parentheses depth:

TreeDiff	Random Masking
for char in group: if char == '(': stack.append(char) max_depth = max(max_depth, len(stack)) elif char == ')': if stack: stack.pop()	count = 0 for char in group: if char == '(': count += 1 # Fails to track # nested peak depth

☆ Key Contributions

- **First AST-aware masking** for DLLMs in code generation
- **13.3% relative improvement** on HumanEval+ vs random masking
- Trained on **150K** long code reasoning samples
- Negligible overhead: AST parsing is O(L)

EvalPlus Robustness



TreeDiff maintains stable EvalPlus Pass@1 across inference steps (T=256→512), while baselines degrade by ~2.1%.

Conclusion

TreeDiff addresses a fundamental limitation: **random masking ignores code structure**. By aligning corruption with AST hierarchy, the model learns programming language compositionality, reconstructing programs that respect grammatical boundaries and capture long-range dependencies.

Impact: Structural priors unlock DLLMs for code — TreeDiff narrows the gap between diffusion and autoregressive paradigms.

Setup: LLaDA-8B backbone · 8 × NVIDIA A100 · LoRA (r=64, q=128) · 4096 max tokens · LR 5×10⁻⁶ · Benchmarks: HumanEval+, MBPP+